

Project Title: Image Classification using CNN for Lung cancer

Phase 2: Reflection Report

1. Introduction

In Phase 2 of the project, the goal was to deploy a Binary Neural Network (BNN) on a PYNQ board for image classification. This phase involved loading overlays, setting up the software environment, and running initial tests to validate the setup. Despite following the GitHub repository and related instructions, several technical challenges arose, primarily due to system compatibility and missing critical libraries.

2. Steps Undertaken Chronologically

1. Environment Setup:

- a. Imported necessary Python libraries: `pynq`, `cffi`, and `bnn`.
- b. Defined the FPGA overlay path and attempted to load the bitstream file `procsys_wrapper.bit`.

2. FPGA Bitstream Download:

- a. Successfully downloaded the overlay onto the FPGA using the `Overlay` class.

3. Interface Setup with C Functions:

- a. Used `cffi` to define the C functions required to communicate with the BNN library.

4. Loading Parameters:

- a. Defined the `load_params` function to load pre-trained model parameters to the FPGA.
- b. Encountered an initial **SyntaxError** due to incorrect function definition syntax.

5. Error - **AttributeError**:

- a. After correcting syntax, faced an **AttributeError**:
 - i. Message: Could not find IP or hierarchy `load_parameters` in overlay
- b. Interpretation: The `load_parameters` function was not exposed correctly by the overlay.

6. Attempt to Load Shared Object Library:

- a. Tried loading the `libbnn.so` shared object using `ffi.dlopen()`.
- b. Encountered an **OSError**:
 - i. Message: cannot load library 'libbnn.so': No such file or directory

7. Analysis and Investigation:

- a. Verified that the bnn repository was cloned properly.
- b. Confirmed that the missing `libbnn.so` library was critical for running inference.
- c. Investigated potential causes: missing compilation step, repository version mismatch, or system environment misconfiguration.

3. Challenges Encountered

Challenge	Description	Impact
SyntaxError	Wrong function syntax used.	Quick fix, minor setback.
Attribute Error	<code>load_parameters</code> not found in overlay.	Major obstacle; indicates a mismatch or incomplete overlay.
OSError	<code>libbnn.so</code> library missing.	Critical failure; prevents running any inference.

4. Root Causes Identified

- **PYNQ Image Version Compatibility:**
 - Currently using **PYNQ 2.6**.
 - Most BNN repositories were designed for **PYNQ 2.3 or 2.4**.
 - API changes in PYNQ 2.6 likely cause missing or broken function links in the overlay.
- **Missing Library (`libbnn.so`):**
 - The GitHub repository may not include pre-compiled libraries.
 - `libbnn.so` might need to be manually compiled from source.
- **Overlay Incompatibility:**
 - The `.bit` file loaded may not contain the correct hardware IPs expected by the software layer.

5. Critical Questions for Progress

- Is my PYNQ board image (2.6) fully compatible with the BNN repository requirements?
- Does the cloned repository include precompiled binaries or must I build them manually?
- Can I manually verify the content of the bitstream to confirm the expected IPs are present?
- Should I downgrade to PYNQ 2.4 for better compatibility?

6. Reflections and Lessons Learned

- **Software and hardware version alignment is critical** in embedded system projects.
- **Documentation verification** at the beginning could save significant troubleshooting time.
- **Error analysis and methodical debugging** are essential skills; quick "trial and error" does not work in FPGA-based systems.
- **Understanding hardware overlays deeply** (i.e., .bit and .hwh files) is necessary.

7. Proposed Next Steps

- Verify the .hwh file to check if load_parameters IP exists.
- Attempt to build libbnn.so manually if source files are available.
- If unsuccessful, downgrade to **PYNQ 2.4**.
- Revalidate the entire setup against the repository's documented environment.

8. Conclusion

Phase 2 provided critical insights into the intricate relationship between hardware overlays, software libraries, and system configurations. Although progress was slowed by technical barriers, it highlighted the importance of system compatibility checks and thorough initial setup. Addressing these questions will be key to progressing into Phase 3, where successful deployment and inference of the BNN model are the objectives.